BASTIONS OF RETROCOMPUTING

A strategy game for the Sharp PC-1500, PC-1500A and Tandy PC-2

Instructions – version 2.2



1. The game

The Cathodic Church (CC) – a fictional organisation with a profound interest in vintage computing – maintains ten places across Europe, called Bastions of Vintage IT, numbered 0 to 9: Lourdes (France), Fatima (Portugal), Altotting (Germany), Santiago de Compostela (Spain), Jasna Gora (Poland), Avila (Spain), Loreto (Italy), Valere (Switzerland), Hohensalzburg (Austria) and the Vatican. In these bastions of retro computing the CC keeps its vintage and NOS treasures: old hardware from IBM, HP, Apple, Commodore, Atari, Sharp, Casio, NEC and others. They are revered and heavily protected, and no outsider is let in. The security systems of every bastion are run by Tech Priests, who also guard its gold and its retro stuff. You lead a small team of Ghosts, who strike with API keys, and Hackers, who run Killer Disks. New volunteers join you as you conquer bastions, and you can hire Gurus, the masters of Dependency Hell, and more Hackers. Conquer all ten places to win the war.

The first places are easy. Every next place is harder than the one before, and the last three, places 7, 8 and 9, are the richest places: they start with twice as many Tech Priests and more than twice as much gold, but not with twice as much retro stuff, so what one of them gives you is not enough to conquer the next one. The richest of all is the Vatican (place 9). Every place also hides its own retro treasure, and every place has a **weak spot**: one of your weapons hurts it more than the others. Once you hold Avila or a higher place, the Tech Priests strike back and your places may rise against you; Hackers you leave behind as a **garrison** can hold them.

2. What you need

- A Sharp **PC-1500A**, or a **PC-1500** or **Tandy PC-2** with a RAM module (CE-151 or larger). The recording holds the program without its REM lines (5702 bytes); with its variables it needs about 5.9 KB of memory, so it does not fit into a PC-1500 or PC-2 without a module.

- To load the recording: the **CE-150** printer/cassette interface (for the PC-2: the Radio Shack printer/cassette interface) and a computer, phone or player that can play WAV files.

3. Loading the game

1. Switch off the computer, insert it into the CE-150 and connect the audio output of the computer or player to the **EAR** jack of the CE-150.
2. Switch on, select PRO mode and type `NEW0` [ENTER], then `CLOAD` [ENTER].
3. Play `bastions-v2.2.wav` (name on the tape: `BASTIONS`, about 7 minutes). If the program does not load, try again with another volume.
4. Select RUN mode and type `RUN` [ENTER].

You can also type in the listing `bastions_of_retrocomputing-v2.2.txt`. Leave out the REM lines: the recording does not contain them either, and with them the program would not leave enough memory for its variables on a PC-1500A. The place names and treasures in lines 970–986 contain small letters (SML key).

4. Start and main menu

The game shows `CONQUER THE BASTIONS` while it prepares the places, then `OF RETROCOMPUTING !` while it plays the opening of J.S. Bach's Toccata in D minor, and the main menu:

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FIGHT/BUY/SELL/RESOUR/END?
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Press one key, without ENTER:

Key	Action
F	Fight: attack a place (section 7)
B	Buy Killer Disks, API keys, Gurus and Hackers (section 8)
S	Sell retro stuff, Hackers and Gurus (section 9)
R	Show your resources (section 10)
E	End the game (section 11)

Other keys are ignored. Numbers (the Killer Disks and API keys for a fight, the garrison, and what you buy or sell) are typed right after the question mark and confirmed with ENTER; any key other than a digit wipes the number, so you can type it again. A number that is too big gives a beep and the question again.

5. Your forces

At the start you have:

Resource	Start
Ghosts	10
Hackers	5
Gurus	0
Killer Disks	20
API keys	4
Retro stuff	0
Gold	2000

Your people

- **GHOSTS** are silent, brilliant programmers who strike with surgical precision: they write unholy code that finds every flaw in the Tech Priests' systems, making years of built-up

security collapse like a house of cards—unleashing glitches, worm plagues, and system destruction vectors to ultimately crack open the “Miracle Exploit”. Ghosts need API keys to attack: in every round each Ghost uses one of the API keys you chose for the fight, until all of them are used. A Ghost without an API key still attacks with its own code, but six times weaker than with a key. You cannot hire Ghosts: they join you when you conquer a place. The Tech Priests reach them only when no Hackers are left.

- **HACKERS** (HACK on the display) run the Killer Disks: in every round each Hacker runs one of the Killer Disks you chose for the fight, until all of them are used, so the more Hackers you have, the faster your disks strike. A Hacker without a Killer Disk still attacks with plain code and viruses, but ten times weaker. Hackers also stand in the front line: the Tech Priests strike them first. Every conquered place gives you new Hackers, and you can also hire them.
- **GURUS** fight alone, without Killer Disks or API keys, in every fight, and stay with you until they are eliminated. They strike with a huge, carefully prepared Dependency Hell, which paralyses the Tech Priests’ systems and ends in crashes or a thermal infernal meltdown: a Guru who strikes eliminates 12 Tech Priests. Groups of seven Tech Priests hunt them, and a Guru who faces more than 10 Tech Priests at once may fall.

Your people are Ghosts + Hackers + Gurus. When all of them are gone, the Cathodic Church (CC) wins.

Nobody gets hurt in this war. It is fought with code, not with swords. When a Hacker, a Ghost or a Guru is “eliminated”, the Tech Priests have defeated their solution, their code or their system: a patch closes the hole, a firewall swallows the worm, a Killer Disk is wiped clean. The defeated hero leaves the battlefield in one piece, for a long relaxation and a complete rewrite. Your people eliminate Tech Priests the same way: their systems crash, and they spend the rest of the war in the Cathodic Church’s retreat centre, waiting for the main spiritual help desk.

Weapons

- **Killer Disk** (`K-DISKS` on the display): a disk on which the Hackers prepare killer programs – malevolent system daemons, endless loops, merge conflicts, malicious code and deadlocks – that drive the Tech Priests’ machines into an infernal overheat or a thermal meltdown, and the Tech Priests into panic. Only Hackers can run Killer Disks, each Hacker one disk per round. Every Killer Disk is used only once; a disk that hits eliminates one Tech Priest.
- **API key:** the key that lets a Ghost into the Tech Priests’ systems. Only Ghosts can use API keys, each Ghost one key per round. Every API key is used only once; a Ghost’s attack that hits eliminates three Tech Priests.

Not every attack succeeds. You choose how many Killer Disks and API keys to take into each fight. Each of them is used only once: in every round each Hacker runs one Killer Disk and each Ghost uses one API key, as long as any are left. When you win, the Killer Disks and API keys that were not used come back to you.

6. The places

At the start of the game the places hold:

Place	Name	Tech Priests	Retro stuff	Gold
0	Lourdes	13	5	400
1	Fatima	22	9	300
2	Altotting	33	13	600
3	Santiago de C.	50	17	900
4	Jasna Gora	75	21	1200
5	Avila	115	25	1500
6	Loreto	182	29	1800
7	Valere	588	33	4900
8	Hohensalzburg	976	37	5600
9	Vatican	1646	41	6300

Places 7, 8 and 9 (shaded) are the richest places.

Weak spots. Every place is weak to one of your weapons: Killer Disks, API keys or Gurus. When you choose a place, the game shows it, e.g. `WEAK TO KEYS`. There, the hits of that weapon count one and a half times. The weak spots follow the places in a fixed order – disks, keys, Gurus, disks, keys … – but where the order starts is drawn anew in every game: once you know the weak spot of one place, you know them all. Plan your purchases for the next places.

7. Fight

1. Press F. The game asks `WHICH PLACE 0-9 (L=LIST)?`. Press the number of the place. L shows all places, one per second, e.g. `2 Altotting: 25 TP` (Tech Priests still there), `0 Lourdes: YOURS` or `6 Loreto: 10 GAR` (yours, with a garrison of 10 Hackers). Any other key returns to the main menu.
2. The place is shown, two seconds per screen: `PLACE 3: Santiago de C.`, `50 TECH PRIESTS`, `17 RETRO STUFF, 900 GOLD` and its weak spot, e.g. `WEAK TO K-DISKS` (`K-DISKS`, `KEYS` or `GURUS`). A place that you have conquered before and that broke free gives only a fifth of its gold and of its retro stuff; the screen already shows the reduced amounts. If the place is already yours, the game shows e.g. `Lourdes IS YOURS` and asks for the place again.
3. `K-DISKS (MAX 20) ?` and `API KEYS (MAX 4) ?`: the game shows how many you have. Type the number right after the question mark, as when buying, and press ENTER (ENTER alone: none). More than you have gives a beep and the question again. With no Killer Disks and API keys at all, your Hackers and Ghosts fight with their own code only, and much weaker (see Rounds).
4. `ENTER=FIGHT 3=RETREAT`: press 3 to retreat (you get your Killer Disks and API keys back) or ENTER to fight.
5. First `FIGHTING!` appears in the middle of the display; then your team and the Tech Priests march twice from both sides of the display, and each time they meet, the game beeps. Then the rounds begin.

Rounds

The fight is played in rounds:

- Your side strikes first: the Killer Disks your Hackers run (one per Hacker, while disks are left), the API keys your Ghosts use (one per Ghost, while keys are left) and all your Gurus. Hackers and Ghosts without a weapon strike too, but weaker: a Hacker without a disk ten times, a Ghost without a key six times. At the place's weak spot, the hits of that weapon count one and a half times (not for the Hackers and Ghosts without a weapon). The first strike is the weakest, because the Tech Priests are still behind their firewalls; from the second round on your attacks hit more often. Killer Disks are the exception: their killer programs get through the firewalls and hit just as often in the first round. However much you attack with, at most a third of the Tech Priests the place had at the start of the fight fall in one strike: the others take cover, and what was used on them is lost. So the Tech Priests are eliminated step by step, in three strikes or more.
- Then the Tech Priests who are still active strike back. Your Hackers stand in the front line: the Tech Priests eliminate Hackers, and only when no Hackers are left they turn on the Ghosts. Groups of Tech Priests can also eliminate Gurus, and when after your Killer Disks and API keys more than 10 Tech Priests per Guru are left, each Guru may also fall.
- On the display, the game first shows `ROUND1:CC m vs YOU p` with both sides at full strength: the Tech Priests of the place and all your people. After every strike the next round (`ROUND2`, `ROUND3` …) shows what is left: the Tech Priests still active after your strike and your people still active after theirs. The last round shows `CC 0` when you win or `YOU 0` when you lose. After every strike in which your Hackers run Killer Disks, the game shows `Hackers launch K-DISKS!`, after every strike in which your Ghosts use API keys, it shows `Ghosts cast worms plague!`, and after every strike in which your Gurus attack, it shows `Gurus`

`set Dependency Hell`, before the next round. Each of these messages comes with the last three notes of the second phrase of Bach's Toccata. When the Tech Priests' blows eliminate your last Hackers, the game shows `NO HACKERS LEFT!`, and when they eliminate your last Ghosts, `NO GHOSTS LEFT!`. Before every new screen of a round the display goes blank for a moment, so that you see each change.

How your forces are used. In the first round each of your Hackers runs one of the Killer Disks you took and each Ghost uses one API key: if you took no more of them than you have Hackers and Ghosts, they are all used at once, in the weakest strike. Every disk and key is used only once. When they are used up, your Hackers and Ghosts do not just stand there as shields: they fight on with their own code and viruses, a Hacker ten times and a Ghost six times weaker than with a weapon. The Tech Priests strike your Hackers first, and in most fights the last of them fall in the first round or two: `NO HACKERS LEFT!`, and the disks still in the fight are not used any more. Then your Ghosts are hit; after `NO GHOSTS LEFT!` the keys left lie idle too, and only your Gurus fight on. The fight goes on until no Tech Priests or none of your people are left; once it has begun, only surrendering stops it (see Surrender below). Disks and keys that were not used come back to you when you win or surrender.

You won

A triumph melody plays while the first screen shows e.g. `YOU WON! (Lourdes)`. The line stays about 1.5 seconds after the melody, and the next screens follow by themselves, about 1.5 seconds each:

`YOU GOT g GOLD`
`r RETRO c K-DISKS`

- You got all the gold and all the retro stuff of the place (only a fifth of them when you win back a place that broke free), and some Killer Disks (more at the higher places), which the Tech Priests left behind!

`LOST:mH sG jU`

- Your losses in this fight, e.g. `LOST:7H 2G 1U` (H Hackers, G Ghosts, U Gurus).

`+y HACKERS, +z GHOSTS`

- A sixth of the Tech Priests of the place defect to you as Hackers. The more gold a place gives you, the more new Ghosts join you, because the IT people were outraged by how much gold the CC had hoarded there.

`AND ALSO YOU GOT:`

- Last, the retro treasure of the place, e.g. `Casio FX-602P` at Lourdes. Every place has its own treasure; it is a trophy and does not change your resources. You get it only the first time you conquer the place: when you win back a place that broke free, this screen is left out. The treasure stays on the display for about 3 seconds.

The place is now yours. Last, unless it was your last place, the game asks `GARRISON (MAX n) ?`: type how many of your Hackers stay in the place as its garrison and press ENTER (ENTER alone: none). They leave your army for good: you cannot take them back, sell them or use them in a fight. A garrison guards the place when it rises (see A place breaks free). The question is left out when you have no Hacker to spare.

Surrender

From the second round on, while you have fewer people than the Tech Priests still active, `S` appears at the right end of the round screen, e.g. `ROUND3:CC 120 vs YOU 45  S`. Press S while that screen is on the display (hold the key until the screen changes) and you surrender: `YOU SURRENDERED!` appears in the middle of the display while the falling melody plays, and blinks three times; then your losses in this fight are shown, `LOST:mH sG jU`, and the main menu comes back.

You keep the people you still have and the Killer Disks and API keys that were not used; you get no gold, no retro stuff and no new people. The place stays with the CC, but only with the Tech Priests still active: the ones you have eliminated do not come back, so your next attack on that

place is easier. When you have as many people as the Tech Priests or more, there is no S: fight on!

You lost

`YOU LOST!` appears in the middle of the display while a falling melody plays, and then it blinks three times. A fight is lost only when you have nobody left, so a lost fight always means `NO ONE LEFT!` and `GAME OVER! Priests won!`. Then, two seconds later, it shows `..but you have got...` and the treasure of the last place you conquered, or `lesson in humility..` if you have not conquered any. Meanwhile the second phrase of Bach's Toccata plays.

A place breaks free

Once you have conquered Avila (place 5) or a higher place, the Tech Priests start to fight back. From time to time, on the way back to the main menu, one of your places rises against you; it is never the place of your last fight. A garrison may hold it: every Hacker of the garrison holds off 40 of the Tech Priests the place had at the start of the game, so the chance is 40 × garrison ÷ Tech Priests, and a garrison of a fortieth of them always holds (Lourdes, Fatima, Altotting: 1 Hacker; Santiago de C., Jasna Gora: 2; Avila: 3; Loreto: 5; Valere: 15; Hohensalzburg: 25). Then the game shows e.g. `Loreto held out!` and the garrison stays.

Otherwise the place breaks free: the game shows e.g. `Loreto broke free!` while the music plays, then the main menu comes back. The garrison is lost, and the CC sends three times as many Tech Priests as the place had at the start of the game. Winning it back gives only a fifth of its gold and of its retro stuff, and no treasure: a place that breaks free is always a loss. To win the war, all ten places must be yours at the same time.

8. Buy

Press B. The game shows `D)ISK A)PI G)URU H)ACKER`:

Key	Item	Price (gold)
D	Killer Disk	8
A	API key	20
G	Guru	400
H	Hacker	80

The game shows the price and how many you can afford, e.g. `PRICE 8 (MAX 250) ?`. Type the number on the same line, right after the question mark, e.g. `PRICE 8 (MAX 250) ? 12`, and press ENTER (ENTER alone buys nothing). Any key other than a digit wipes the number, so you can type it again; more than you can afford gives a beep and the question again. If you cannot afford even one, the game beeps, shows `NOT ENOUGH GOLD` for a second and the buy menu comes back at once. After a purchase the buy menu comes back too: buy something else or press any other key to return to the main menu.

9. Sell

Press S. The game first shows what you have to sell, e.g. `RET 9 HACK 12 GURU 2` (retro stuff, Hackers, Gurus) and waits until you press any key. Then it asks `SELL R)ETRO H)ACKER G)URU?`:

Key	Item	Price (gold)
R	Retro stuff	475 at the start
H	Hacker	50
G	Guru	320

Press R, H or G; any other key returns to the main menu. The game shows the price of one piece and how many you have, e.g. `PRICE 320 (MAX 2) ?`. Type the number right after the question mark, as when buying, and press ENTER (ENTER alone sells nothing). Then it shows `GOLD NOW` g for about 1.3 seconds and the sell menu comes back.

The price of retro stuff is 475 gold at the start of the game. After every fight it rises by 0, 50 or 100 gold (at random); every sale of retro stuff lowers it by 0 to 250 gold, but never below 250. On average over a game the price stays level: whoever sells often pushes it down, whoever waits sees it rise. Hackers and Gurus sell for less than they cost. You cannot sell your last people: the game shows `KEEP SOMEONE!`. If you have none of the chosen item, it shows `NOTHING TO SELL`.

10. Resources

Press R. The game shows `YOU HAVE: API KEYS` n, then `K-DISKS` n, your people `GHOST` n `HACK` n `GURU` n and last `GOLD` n `RETROSTUFF` n, one screen at a time. Press any key to see the next screen; after the last one the main menu comes back.

11. End

Press E: the game shows `Y=END`. Press Y to end the game at once, without a message; any other key brings the main menu back.

12. Winning and losing

You win the war when all ten places are yours: `ALL BASTIONS ARE YOURS!` and `YOU BEAT THE TECH PRIESTS!`. The second message appears by itself while the whole opening of Bach's Toccata plays; then the display is cleared and the game ends. You lose when none of your people are left: `GAME OVER! Priests won!`.

13. Tips

- Every element counts. Killer Disks are the cheapest, but only your Hackers can run them, one each per round; API keys need Ghosts, one each per round; Gurus need nobody and fight again in every round and every fight. Without their weapons, Hackers and Ghosts hit far weaker. Every Killer Disk and API key is used only once, and at most a third of the Tech Priests fall in one strike: when you have more Ghosts than API keys, all your keys are used in the first round, while the Tech Priests are still behind their firewalls. A player who leaves out any of Killer Disks, API keys, Gurus, Hackers, Ghosts or garrisons wins clearly less often.
- Your income is the gold and the retro stuff of the conquered places. Retro stuff gains value with every fight and loses some with every sale: keep it while you can, and when you sell, sell a lot at once.
- Save gold and retro stuff from the first places for the rich places 7, 8 and 9: each of them is a big step up, and what one of them gives you is not enough for the next one. If you use up all your gold in every fight, you will not get through Hohensalzburg and the Vatican.
- Gurus are expensive and mighty, and they fight again in every following fight – but they are no miracle workers. Do not let them be outnumbered: when more than 10 Tech Priests per Guru are left after your Killer Disks and API keys, they fall fast.
- Use the weak spots: buy the weapon the next places are weak to, and at a place weak to Gurus let them do the heavy work.
- Garrisons cost Hackers for good, but a few are enough: one Hacker holds off 40 Tech Priests. A place that breaks free comes back with three times as many Tech Priests and gives only a fifth of its gold and retro stuff when you win it back, so guard the bigger places you hold. The first, small places are cheap to win back; a half garrison holds half the time.
- Hackers stand in the front line: as long as you have enough of them, the Tech Priests' blows do not reach your Ghosts, and your API keys keep working.

- Surplus Hackers and Gurus can be sold for gold, but every Hacker you sell is one fewer to run Killer Disks and to take the Tech Priests' blows.
- The Hackers you win are your reserve of people and of disk-runners. Check your forces with R and the places with L before you attack.
- Changed your mind before the fight? Press **3** to retreat: you lose nothing.
- A lost fight ends the game. When a fight goes badly, press S in time (from the second round, while you have fewer people than the Tech Priests): you keep your people and your unused disks and keys, and the Tech Priests you have eliminated are gone for good. Then buy what you need and attack the same place again.